

Project Proposal: Predicting Road Network Failure Probabilities for Post-Disaster Drone Routing

Overview: This project develops a data-driven model to predict the failure probabilities of road segments (arcs) in a transportation network following a disaster. By applying the full data science pipeline, the project transforms raw infrastructure and environmental data into empirical "p" values that represent the likelihood of a road being blocked or damaged.

Motivation: In post-disaster scenarios, optimizing the travel time of drones used to assess network connectivity is critical. Current research models often rely on assumed probabilistic information for arc failures; this project aims to provide these models with realistic, data-backed inputs. Bridging the gap between theoretical optimization and real-world data will significantly enhance the accuracy of disaster response strategies.

Datasets and Data Collection: The project will utilize a hybrid dataset to meet the enrichment requirements:

- **Primary Data:** Road network geometries (nodes and arcs) will be programmatically collected from OpenStreetMap (OSM) using the OSMnx library.
- **Enrichment Data:** To enrich the public data, I will integrate building footprints (extracted via OSM) to calculate structural density and utilize seismic risk maps (PGA values) from the IBB Open Data Portal or AFAD. All data processing and spatial joining will be performed using Python and GeoPandas. The final dataset is expected to include 1,000 to 5,000 road segments from a selected high-risk district.

Hypothesis: I hypothesize that there is a statistically significant positive correlation between local building density and road failure probability in disaster zones. Specifically, road segments located in areas with a higher "building-to-road-length" ratio and higher seismic risk scores will demonstrate a higher predicted probability of being unusable compared to segments in low-density or low-risk areas.